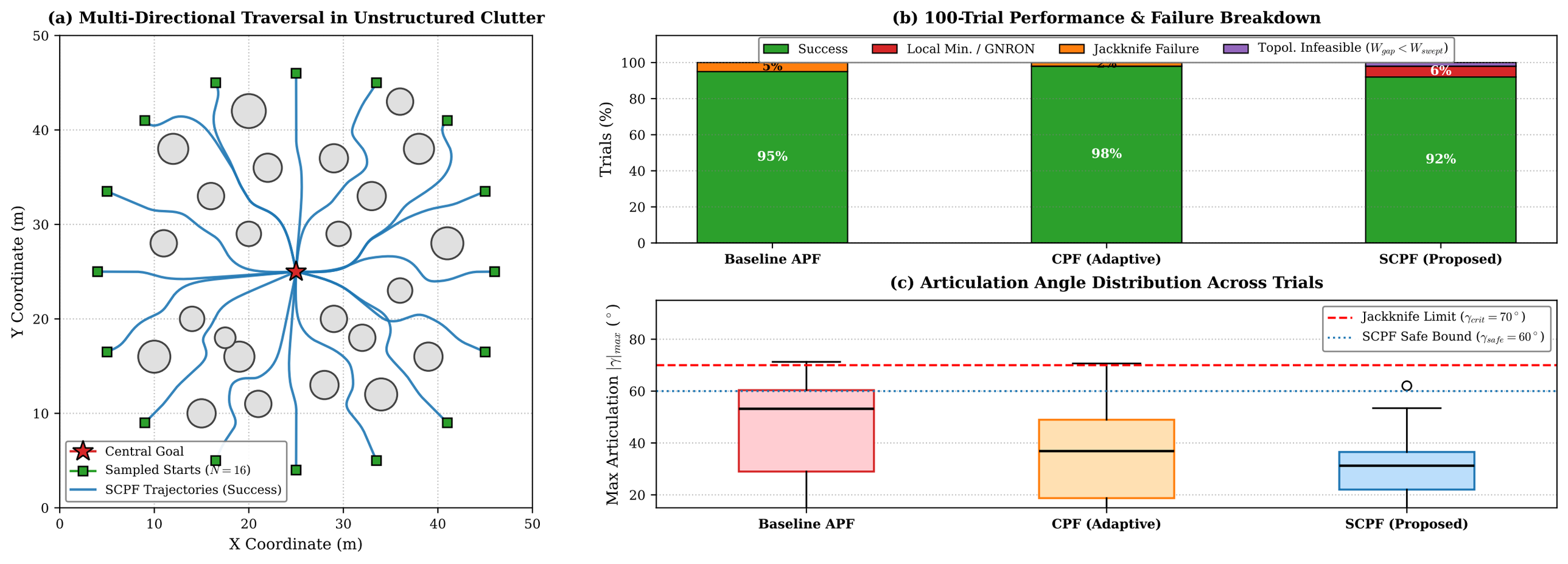

(a) Multi-Directional Traversal in Unstructured Clutter
X Coordinate (m)
Y Coordinate (m)
Central Goal
Sampled Starts ($N = 16$)
SCPF Trajectories (Success)
(b) 100-Trial Performance & Failure Breakdown
Success
Local Min. / GNRON
Jackknife Failure
Topol. Infeasible ($W_{gap} < W_{swept}$)
Trials (%)
95%
98%
92%
6%
Baseline APF
CPF (Adaptive)
SCPF (Proposed)
(c) Articulation Angle Distribution Across Trials
Max Articulation $|\gamma|_{max}$ (°)
Jackknife Limit ($\gamma_{crit} = 70°$)
SCPF Safe Bound ($\gamma_{safe} = 60°$)
Baseline APF
CPF (Adaptive)
SCPF (Proposed)